



Gennoor Tech

TRAIN. INNOVATE. BUILD.

5-DAY DEVOPS + ML ENGINEERING BOOTCAMP · 40 HOURS

Production AI & MLOps

Bridge the gap between PoC and production — at enterprise scale

5

DAYS

40

HOURS

18+

HANDS-ON LABS

3

CLOUD PLATFORMS

Course Overview

This 5-day bootcamp covers the complete MLOps lifecycle — from experiment tracking and model versioning through CI/CD pipelines, containerized deployment, and production monitoring across Azure ML, AWS SageMaker, and open-source tools. Participants design and implement a fully automated ML pipeline from data ingestion to live inference by the end of the program.

85% of ML models never make it to production. This bootcamp exists to change that number for your organization. You'll learn the engineering practices, tooling, and automation patterns that separate experimental notebooks from reliable, scalable, monitored AI systems. Designed for teams that already know how to build models — and now need to ship them.

Who Is This For?



ML Engineers

Engineers deploying and maintaining ML models in production



DevOps Engineers

DevOps teams extending CI/CD to ML workloads



Platform Engineers

Teams building internal ML platforms and infrastructure



Data Scientists

Data scientists learning to productionize their own models

Learning Outcomes

- 1 **Design end-to-end ML pipelines** — data ingestion, feature engineering, training, evaluation, and deployment as automated workflows
- 2 **Implement CI/CD for ML** — automated testing, model validation, and deployment using GitHub Actions and Azure DevOps
- 3 **Deploy models at scale** — containerize with Docker, serve with FastAPI, deploy to Azure ML, AWS SageMaker, and Kubernetes
- 4 **Monitor production models** — detect data drift, model degradation, and performance anomalies with automated alerting
- 5 **Version everything** — track experiments with MLflow, version datasets with DVC, manage model registries across environments
- 6 **Build across clouds** — implement MLOps patterns on Azure ML, AWS SageMaker, and GCP Vertex AI with open-source tooling

Prerequisites

Python programming (intermediate), basic ML model building experience (Bootcamp 4 or equivalent), familiarity with Git and command line. Docker experience is helpful but not required — covered on Day 1.

5-Day Curriculum

Interactive format with live coding, hands-on labs, and project-based learning. Schedule and session order are fully customizable based on your organization's needs and time zones.

01

MLOps Foundations — Experiment Tracking & Versioning

LECTURE + LIVE CODING + HANDS-ON LABS

SESSION A

MLOps Principles & Architecture

- What is MLOps? Maturity levels (0 → 3) and organizational patterns
- ML lifecycle: data → train → validate → deploy → monitor → retrain
- MLOps vs DevOps: what's different about ML systems
- Infrastructure overview: Docker, registries, orchestrators
- Lab: Set up MLOps development environment (Docker, Git, MLflow)

SESSION B

Experiment Tracking & Model Versioning

- MLflow: tracking experiments, parameters, metrics, and artifacts
- MLflow Model Registry: staging, production, archiving
- DVC (Data Version Control) for dataset versioning
- Lab: Track a complete ML experiment lifecycle with MLflow
- Lab: Version datasets and models with DVC + Git

02

Containerization & Model Serving

LIVE CODING + HANDS-ON LABS + ARCHITECTURE WORKSHOP

SESSION A

Docker for ML Engineers

- Docker fundamentals: images, containers, Dockerfiles, volumes
- Multi-stage builds for lean ML containers
- Docker Compose for multi-service ML architectures
- Lab: Containerize an ML model with all dependencies
- Lab: Build a multi-container app (API + model + database)

SESSION B

Model Serving & APIs

- FastAPI for ML inference endpoints
- Batch vs real-time inference patterns
- Model serialization: pickle, joblib, ONNX, TorchScript
- Lab: Build a production inference API with FastAPI + Docker
- Lab: Implement A/B testing with canary deployments

03

CI/CD for Machine Learning

LIVE CODING + HANDS-ON LABS + PIPELINE DESIGN

SESSION A**GitHub Actions for ML Pipelines**

- CI/CD principles applied to ML: test → validate → deploy
- GitHub Actions: workflows, triggers, secrets, artifacts
- Automated model testing: unit tests, integration tests, data validation
- Lab: Build a GitHub Actions pipeline that trains, tests, and registers a model
- Lab: Automated model quality gates (accuracy thresholds, drift checks)

SESSION B**Azure DevOps & ML Pipelines**

- Azure DevOps Pipelines for ML workloads
- Azure ML Pipelines: components, datasets, compute targets
- Multi-environment deployment: dev → staging → production
- Lab: Build an Azure DevOps pipeline with model promotion
- Lab: Implement automated retraining on data drift detection

04

Cloud MLOps — Azure, AWS & GCP

LIVE DEMOS + HANDS-ON LABS + MULTI-CLOUD WORKSHOP

SESSION A**Azure ML & AWS SageMaker**

- Azure ML: workspaces, compute, endpoints, managed online inference
- Azure ML Pipelines with components and designer
- AWS SageMaker: training jobs, model registry, endpoints
- Lab: Deploy a model to Azure ML managed endpoint
- Lab: Deploy the same model to SageMaker endpoint

SESSION B**GCP Vertex AI & Multi-Cloud Patterns**

- GCP Vertex AI: pipelines, model registry, prediction
- Comparing cloud platforms: cost, features, lock-in tradeoffs
- Open-source MLOps stack: MLflow + Airflow + Kubernetes
- Lab: Build a cloud-agnostic ML pipeline with MLflow
- Lab: Kubernetes deployment with Helm charts for ML

05

Monitoring, Governance & Capstone

CAPSTONE PROJECT + ARCHITECTURE REVIEW + PRESENTATIONS

SESSION A**Production Monitoring & Observability**

- Model monitoring: accuracy drift, data drift, concept drift
- Evidently AI and Weights & Biases for monitoring
- Infrastructure monitoring: latency, throughput, error rates
- Lab: Build a monitoring dashboard with drift detection alerts
- Automated retraining triggers and feedback loops

SESSION B**Capstone: End-to-End MLOps Pipeline**

- Team project: Design and implement a complete MLOps pipeline
- Data ingestion → training → validation → deployment → monitoring
- Architecture review: scalability, cost, security, compliance
- Team presentations and expert feedback
- Certification paths and next steps

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AI developments.

Tools & Platforms Covered

All tools used hands-on with live coding and lab exercises

MLflow

DVC

Docker

Kubernetes

GitHub Actions

Azure DevOps

Azure ML

AWS SageMaker

GCP Vertex AI

FastAPI

Evidently AI

Weights & Biases

ONNX

Helm

Python

Git

What You Take Home

**MLOps Pipeline Templates**
Production-ready CI/CD pipeline configs for GitHub Actions and Azure DevOps — train, test, register, deploy, monitor

**Docker Starter Kit**
Optimized Dockerfiles and Compose configs for ML inference services, multi-stage builds, and GPU support

**Monitoring Dashboard**
Pre-built Evidently AI monitoring templates for data drift, model performance, and automated alerting

**MLOps Maturity Assessment**
Framework to evaluate your organization's MLOps maturity level with a roadmap for advancing to Level 3

Certification of Completion

**Gennoor Tech — Certificate of Completion**
Upon completing all 5 days (40 hours), each participant receives a **Gennoor Tech Certificate of Completion** in "Production AI & MLOps". Digitally signed by the Founder & Director. Validates proficiency in ML pipelines, CI/CD for ML, containerized deployment, multi-cloud MLOps, and production monitoring.

Recommended Next Certifications

Microsoft AZ-400
DevOps Engineer Expert

Microsoft AI-300
Machine Learning Operations (MLOps) Engineer Associate

AWS MLS-C01
AWS Machine Learning Specialty

Google PMLE
Professional ML Engineer

Designed & Delivered By

Jalal Ahmed Khan
Founder & Director, Gennoor Tech Private Limited
Senior AI Consultant and **Microsoft Certified Trainer (MCT)** with 14+ years of experience. Led RFP Automation PoC at Tech Mahindra achieving 60% reduction in response time. Established PoC-to-Pilot governance frameworks accelerating client AI adoption by 40%. Built and deployed production MCP servers and CI/CD pipelines for ML workloads.
16+ active Microsoft certifications including AZ-400 (DevOps Engineer Expert), AI-300, AI-103, PL-400, and GitHub certifications. Trained teams at Microsoft, IBM, Boeing, TCS, Capgemini, and 80+ organizations. **M.E. Mechanical Engineering**, Mumbai University.

Delivery Options

 **On-Site** — customized at your office |  **Virtual Live** — interactive via Teams/Zoom |  **International** — GCC, Africa, APAC, UK



Ready to Ship AI to Production?

Contact us to discuss a customized bootcamp
for your organization or register for the next open batch.

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